

Data Analyst — Home Assignment

Analytics Department · 2026

The Scenario

*“We just closed Q1 and I need to present to leadership next week. Before I go in, I need an honest answer: **are we growing in a healthy way, or are we just doing more volume with the same problems?** Dig into the data. Tell me what’s working, what’s broken, and what I should worry about. Come ready to present and defend your findings.”*

What You Have

- **orders.csv** — order-level data from HAAT’s platform
- **payment_methods.csv** — payment instruments registered by customers
- **areas.csv** — area name lookup table

What You Deliver

- A slide deck with your analysis and findings
- Written answers to Q1–Q8 (required)
- Q9 is the presentation. Q10 is a bonus question.

How You Will Be Evaluated

- You will present live, followed by Q&A;
- You will be asked to walk through any number on any slide and explain how you got it
- We care about clear thinking and correct numbers, not formatting
- You may use any tool — SQL, Python, Excel, or BI tools

Note: There are no right or wrong answers for most questions. What matters is your business understanding, analytical thinking, and your ability to explain your reasoning. We are looking for how you think, not just what you find.

Dataset Schema

orders.csv

One row per order placed on the platform.

Column	Type	Description
Id	INT	Unique order identifier
OrderDate	TIMESTAMP	When the order was placed
Areald	INT	Area where the order was placed — join to areas.csv for name
BusinessId	INT	The merchant who received the order
DriverId	INT	Courier assigned. NULL if never assigned
UserId	INT	Customer who placed the order
Price	DECIMAL	Order value in ILS
PaymentMethod	INT	0 = Cash, 1 = Credit, >1 = Other
RejectedDateByRestaurant	TIMESTAMP	When merchant rejected. NULL if not rejected
ArriveDate	TIMESTAMP	When courier arrived at customer. NULL if undelivered
DriverCandidateAssignedDate	TIMESTAMP	When courier was first assigned. NULL if never assigned
PickedDate	TIMESTAMP	When courier picked up from merchant. NULL if not picked up
StartPrepare	TIMESTAMP	When merchant started preparing. NULL if not started

payment_methods.csv

One row per payment instrument registered by a customer.

Column	Type	Description
Id	INT	Unique payment method record
UserId	INT	Customer who registered this payment method
Token	VARCHAR	Tokenised identifier for the payment instrument

areas.csv

Lookup table — area names for each Areald.

Column	Type	Description
Areald	INT	Matches Areald in orders table
AreaName	VARCHAR	Display name of the operating area

Assignment Questions

Q1–Q8 are required. Q9 is the presentation. Q10 is a bonus.

Part 1 — Know the Business

Q1. First Look

Explore the dataset and write a short summary of what you see.

Cover the following:

- How many orders are in the dataset? How are they spread across areas and months?
- What is the average order value? What is the most common payment method?
- How long does it typically take from when an order is placed (OrderDate) to when a courier is assigned (DriverCandidateAssignedDate)?
- What is one thing in this data that looks unusual or worth investigating further? Be specific — name the column, the area, or the timeframe.

Q2. Define What Good Looks Like

HAAT has no KPI framework yet. You are building it from scratch.

Define the 5 most important metrics you would track for a delivery platform like HAAT. For each metric:

- Write the exact formula using column names from the dataset
- Explain in one sentence what it measures and why it matters
- Give an example of a healthy value and a concerning value
- Name which team should own it (Operations, Finance, Marketing, etc.)

At least one metric must measure delivery success rate, one must measure how quickly couriers get assigned after an order is placed, and one must measure commercial performance.

Part 2 — Operational Depth

Q3. Delivery Reliability

Analyse delivery performance using OrderDate, ArriveDate, DriverCandidateAssignedDate, and RejectedDateByRestaurant.

- What percentage of orders were successfully delivered (ArriveDate is not NULL)? Break this down by area and by month.
 - Break down order volume and delivery success rate by hour of the day. Do you see any pattern? If yes, describe it — which hours, what changes, and how large is the effect.
 - Compare delivery performance across all areas. Do any areas stand out as clearly different from the rest? What do you think is the reason?
-